MI-28: a determinant inequality for all nonnegative base powers

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For positive definite matrices A, B, every $k \geq 0$, and $0 \leq p \leq 2$, we prove

$$\det(A^k + |AB|^p) \geq \det(A^k + A^p B^p).$$

In fact, a log-majorization of the corresponding normalized matrices holds. The previously proved range $k \geq 2$ is due to Ghabries, Abbas, Mourad, and Assi. We treat the remaining base powers using a normalized order implication. Two applications of the Furuta inequality establish complementary parameter regions. The identity $||AB|^{-1}B| = A^{-1}$ interchanges the two parameters and closes the remaining range.

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Statement and notation

Throughout, $A, B \in \mathbb{C}^{n\times n}$ are positive definite, matrix powers are defined by spectral functional calculus, and $\|\cdot\|$ denotes the operator norm. For a matrix M, write $|M| = (M^*M)^{1/2}$. Eigenvalues of a positive definite matrix are listed in decreasing order. For positive vectors $u, v \in (0,\infty)^n$, the notation $u \prec_{\log} v$ means

$$\prod_{i=1}^{j} u_i \leq \prod_{i=1}^{j} v_i \quad (1 \leq j < n), \qquad \prod_{i=1}^{n} u_i = \prod_{i=1}^{n} v_i.$$

Theorem 1 — The full determinant and log-majorization inequalities

For every $A, B > 0$, $k \geq 0$, and $0 \leq p \leq 2$,

$$\lambda\big(A^{(p-k)/2} B^p A^{(p-k)/2}\big) \prec_{\log} \lambda\big(A^{-k/2} |AB|^p A^{-k/2}\big).$$

Consequently,

$$\det(A^k + |AB|^p) \geq \det(A^k + A^p B^p).$$

The determinant inequality is the positive definite formulation of MI-28, originating in Ghabries's thesis, §2.5 and final Open Problems, Problem 1, pp. 109–110. Although $A^k + A^p B^p$ need not be Hermitian, its determinant is positive: after factoring out A^k, the remaining product $A^{p-k} B^p$ is similar to the positive definite matrix on the left of the log-majorization. The range $k \geq 2$, $0 \leq p \leq 2$ follows already from Ghabries–Abbas–Mourad–Assi, Lemma 2.5; the precise substitution appears in the proof of Theorem 1 below.

Two order implications

We use the Löwner–Heinz inequality

$$0 < X \leq Y, \quad 0 \leq t \leq 1 \quad \Longrightarrow \quad X^t \leq Y^t,$$

and the following convention for the Furuta inequality: if $X \geq Y > 0$, $r \geq 0$, $a \geq 0$, $q \geq 1$, and $(1+r)q \geq a+r$, then

$$(X^{r/2} Y^a X^{r/2})^{1/q} \leq X^{(a+r)/q}.$$

Furuta's original paper is listed in the references. Both facts are also restated in Tanahashi, Propositions 1–2. The r used here is twice the sandwich exponent in that reference.

For $k, p > 0$, let $\mathcal{I}(k,p)$ denote the assertion that, for every $A, B > 0$, with $D = |AB|$,

$$D^p \leq A^k \quad \Longrightarrow \quad B^p \leq A^{k-p}.$$

This notation describes a universal order implication. It makes no assumption that A and B commute.

Lemma 2 — The lower-power Furuta implication

If $k > 0$ and $k/(k+1) \leq p \leq 1$, then $\mathcal{I}(k,p)$ holds.

Proof. Assume $D^p \leq A^k$. In the Furuta inequality, choose

$$X = A^k, \qquad Y = D^p, \qquad r = \frac{2}{k}, \qquad a = \frac{2}{p}, \qquad q = 2.$$

The condition $(1+r)q \geq a+r$ is exactly $p \geq k/(k+1)$. Thus

$$(AD^2A)^{1/2} \leq A^{1+k/p}.$$

Since $D^2 = BA^2B$, we have $AD^2A = (ABA)^2$, and its positive square root is ABA. Congruence by A^{-1} gives $B \leq A^{k/p-1}$. Applying Löwner–Heinz with exponent $p \in (0,1]$ proves $B^p \leq A^{k-p}$. □

Lemma 3 — The higher-power Furuta implication

If $k > 0$ and $1 \leq p \leq 2$, then $\mathcal{I}(k,p)$ holds.

Proof. Assume $D^p \leq A^k$, and set

$$C = ABA, \qquad s = \frac{p(k+2)}{k+p}.$$

The hypotheses give $1 \leq p \leq s \leq 2$. Apply the Furuta inequality with

$$X = A^k, \qquad Y = D^p, \qquad r = \frac{2}{k}, \qquad a = \frac{2}{p}, \qquad q = \frac{2}{s}.$$

Here $q \geq 1$ and $(1+r)q = a+r$. Using $AD^2A = C^2$, we obtain

$$C^s = (AD^2A)^{s/2} \leq A^{k+2}.$$

Because $0 < 2/(k+2) \leq 1$, Löwner–Heinz followed by inversion gives

$$A^{-2} \leq C^{-2s/(k+2)} = C^{-2p/(k+p)}.$$

For every invertible S and real p, a singular-value decomposition gives

$$(SS^*)^p = S(S^*S)^{p-1}S^*.$$

Apply this identity to $S = A^{-1}C^{1/2}$, for which $SS^* = B$:

$$B^p = A^{-1}C^{1/2}(C^{1/2}A^{-2}C^{1/2})^{p-1}C^{1/2}A^{-1}.$$

Congruence in the inequality for A^{-2}, followed by Löwner–Heinz with $0 \le p-1 \le 1$, therefore yields

$$\begin{aligned} B^p &\le A^{-1}C^{\,1+(p-1)(k-p)/(k+p)}A^{-1} \\ &= A^{-1}C^{\,p(k-p+2)/(k+p)}A^{-1}. \end{aligned}$$

The exponent of C in the last line equals $s\theta$, where

$$\theta = \frac{k-p+2}{k+2} \in [0,1].$$

Applying Löwner–Heinz to $C^s \le A^{k+2}$ gives $C^{s\theta} \le A^{k-p+2}$. Substitute this in the bound for B^p to conclude $B^p \le A^{k-p}$. □

Remark 4 — Relation to the negative-power Furuta inequality. The argument after $C^s \le A^{k+2}$ is a direct proof of the needed specialization of the negative-power Furuta inequality. In the convention used here, the relevant parameters in Tanahashi, Theorem 3 are

$$X = A^{k+2}, \quad Y = C^s, \quad a = \frac{k+p}{p(k+2)}, \quad q = \frac{1}{p}, \quad r = -\frac{2}{k+2}.$$

The displayed proof verifies that specialization using only Löwner–Heinz and the singular-value identity.

Interchanging the parameters

Lemma 5 — The parameter swap

Let $0 < p \le k$. If $\mathcal{J}(p,k)$ holds, then $\mathcal{J}(k,p)$ holds.

Proof. Assume $D^p \le A^k$, where $D = |AB|$, and define

$$\widetilde{A} = D^{-1}, \qquad \widetilde{B} = B.$$

The order of the factors is essential. Directly,

$$|\widetilde{A}\widetilde{B}|^2 = BD^{-2}B = B(B^{-1}A^{-2}B^{-1})B = A^{-2},$$

so $|\widetilde{A}\widetilde{B}| = A^{-1}$. Inverting the assumed order gives

$$|\widetilde{A}\widetilde{B}|^k = A^{-k} \le D^{-p} = \widetilde{A}^p.$$

Apply $\mathcal{J}(p,k)$ to this pair:

$$B^k \le \widetilde{A}^{p-k} = D^{k-p}.$$

Since $0 < p/k \le 1$, Löwner–Heinz applied to this inequality gives

$$B^p \le D^{p(k-p)/k}.$$

Since $0 \le (k-p)/k \le 1$, applying it also to $D^p \le A^k$ gives

$$D^{p(k-p)/k} \le A^{k-p}.$$

Together these inequalities prove $\mathcal{I}(k,p)$. If $p = k$, the second exponent is zero and its inequality is simply $I \le I$; the argument still applies. □

Corollary 6 — All base powers up to two

For $0 < k \le 2$ and $0 < p \le 2$, the implication $\mathcal{I}(k,p)$ holds.

Proof. If $p \ge 1$, use Lemma 3. Suppose $0 < p < 1$. If $p \ge k/(k+1)$, use Lemma 2. In the remaining case,

$$p < \frac{k}{k+1} < k \le 2.$$

We establish the swapped implication $\mathcal{I}(p,k)$. When $k \le 1$, its exponent k satisfies

$$\frac{p}{p+1} < p < k \le 1,$$

so Lemma 2, with the parameters interchanged, applies. When $1 \le k \le 2$, Lemma 3, with the parameters interchanged, applies. Lemma 5 now proves $\mathcal{I}(k,p)$. □

Log-majorization and the determinant

Proof of Theorem 1. First take $0 < k \le 2$, $0 < p \le 2$, and write

$$H = A^{(p-k)/2} B^p A^{(p-k)/2}, \qquad Z = A^{-k/2} D^p A^{-k/2}.$$

Both matrices are homogeneous of degree p in B. Let $c = \|Z\| > 0$, and replace B by $c^{-1/p}B$. The resulting Z is at most I, equivalently $D^p \le A^k$ for the scaled pair. Corollary 6 gives $B^p \le A^{k-p}$. Congruence by $A^{(p-k)/2}$ shows that the scaled H is at most I. Scaling back proves

$$\|H\| \le \|Z\|.$$

For every $1 \le j \le n$, apply this norm inequality to $\bigwedge^j A$ and $\bigwedge^j B$. These exterior powers are positive definite and preserve products, inverses, and spectral powers. In particular,

$$\left|(\bigwedge^j A)(\bigwedge^j B)\right| = \bigwedge^j |AB|.$$

The operator norm of an exterior power of a positive definite matrix is the product of its j largest eigenvalues. We thus obtain all partial-product inequalities in the claimed log-majorization. The full determinants agree:

$$\det H = \det Z = \det(A)^{p-k} \det(B)^p.$$

This proves the log-majorization for $0 < k \le 2$, $0 < p \le 2$.

For $k \ge 2$, use the previously established result Ghabries–Abbas–Mourad–Assi, Lemma 2.5. In its notation, for $0 \le t \le s \le K$,

$$\lambda(X^{K-t}Y^t) \prec_{\text{wlog}} \lambda\big(X^{K/2}(Y^{s/2}X^{-s}Y^{s/2})^{t/s}X^{K/2}\big).$$

Take $X = A^{-1}$, $Y = B$, $K = k$, $s = 2$, and $t = p$. The two eigenvalue lists are exactly those of H and Z; the product $A^{p-k}B^p$ is similar to H. Equality of the full determinants changes this weak log-majorization into the claimed log-majorization. This invokes the original range $0 \le t \le s \le K$ without altering any of its parameter restrictions.

When $p = 0$, the two matrices in the claimed log-majorization are both A^{-k}, so equality holds. For $k = 0$, $p > 0$, let $k \downarrow 0$ in the established range $0 < k \le 2$. Functional calculus and the ordered eigenvalues are continuous on the positive definite cone, and each partial-product inequality and the equality of the full determinants persists. Thus the log-majorization holds at every stated endpoint.

Finally, log-majorization is ordinary majorization of the logarithms of the eigenvalues. Applying the convex function $t \mapsto \log(1 + e^t)$ gives

$$\det(I + H) \le \det(I + Z).$$

Indeed, the corresponding inequality for the sums of this convex function is the standard convex-function characterization of majorization. Since $A^{p-k}B^p$ is similar to H,

$$\det(A^k + A^p B^p) = \det(A^k)\det(I + H),$$
$$\det(A^k + D^p) = \det(A^k)\det(I + Z).$$

Multiplying by the positive number $\det(A^k)$ proves the determinant inequality. □

Remark 7 — Scope. The theorem treats every dimension and every parameter pair in the canonical positive definite statement of MI-28. For $k > 0$ and $p > 0$, the positive semidefinite extension follows by applying the determinant inequality to $A + \varepsilon I$ and $B + \varepsilon I$, then letting $\varepsilon \downarrow 0$. The positive definite formulation avoids conventions for zeroth powers of singular matrices.

References

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